

SN 2026kie — Template Subtraction Photometry Report

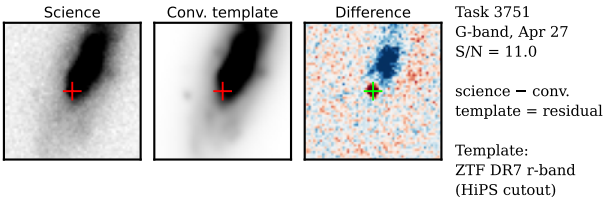
2 May 2026

RAPAS ProAm Network · Gaia G / BP / RP filters · Template subtraction photometry

The RAPAS group (Réseau Amateur-Professionnel pour l'Astronomie des Supernovae, Observatoire de Paris / IMCCE) monitors supernovae and transients through coordinated multi-band photometric follow-up between amateur and professional observatories.

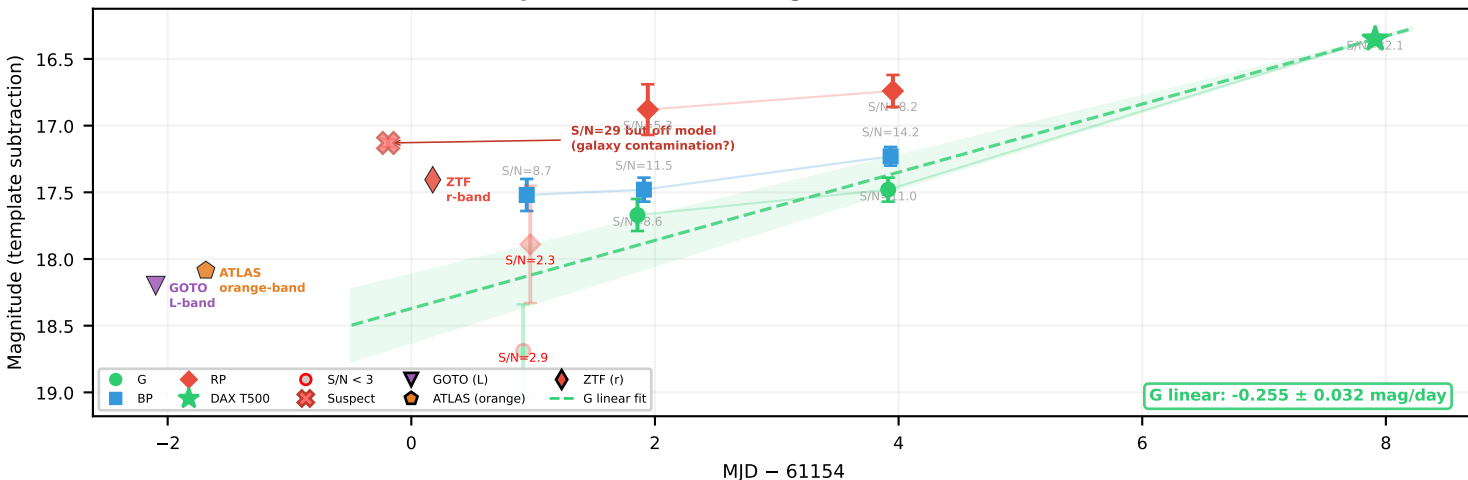
Photometric calibration and template subtraction are performed using STDWeb, a web service built on STDPipe (Karpov, S., 2021, <https://github.com/karpov-sv/stdpipe>), which provides automated astrometry, photometric calibration against Gaia eDR3, and image subtraction using HOTPANTS with ZTF DR7 HiPS templates. For SN 2026kie, template subtraction is essential: the supernova lies on the host galaxy disk, making direct aperture photometry unreliable due to galaxy background contamination.

Template Subtraction Configuration

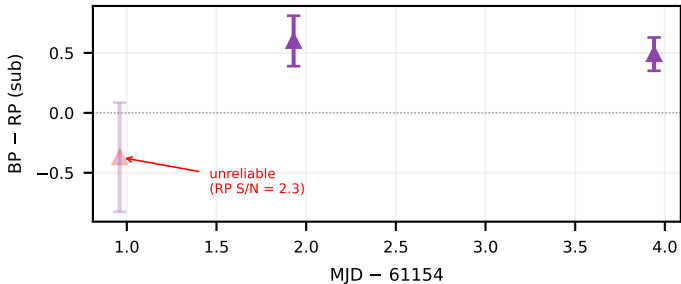


Filter	Calib. mag	Template source	Catalogue	Color term
G	Gaia Gmag	ZTF DR7 r (HiPS)	Gaia eDR3	-0.25
BP	Gaia BPmag	ZTF DR7 g (HiPS)	Gaia eDR3	+0.03
RP	Gaia RPs mag	ZTF DR7 i (HiPS)	Gaia eDR3	-0.18

Template Subtraction Lightcurve — SN 2026kie



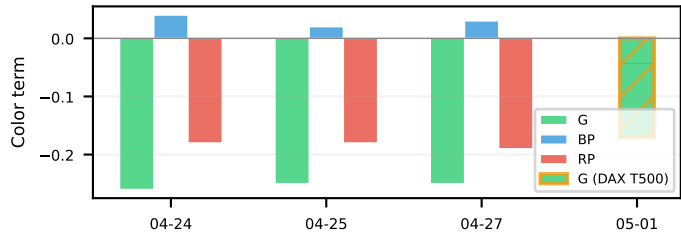
Color Index (subtraction)



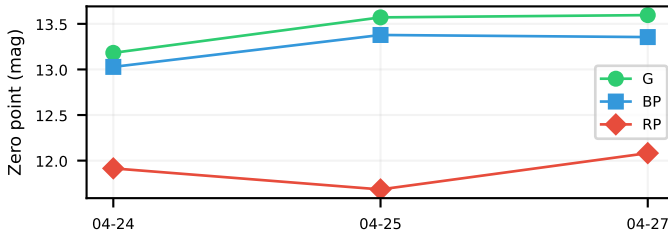
Photometry Summary (all sources)

Date (UT)	MJD	Band	Mag	Err	S/N	Source
04-21.90	61151.90	L	18.20	—	—	GOTO
04-22.31	61152.31	orange	18.09	—	—	ATLAS
04-23.81	61153.81	G	17.13	0.03	29.5	RAPAS
04-24.18	61154.18	r	17.41	—	—	ZTF
04-24.92	61154.92	G	18.69	0.35	2.9	OBS-BZH
04-24.95	61154.95	BP	17.52	0.12	8.7	OBS-BZH
04-24.98	61154.98	RP	17.89	0.44	2.3	OBS-BZH
04-25.86	61155.86	G	17.67	0.12	8.6	OBS-BZH
04-25.91	61155.91	BP	17.48	0.09	11.5	OBS-BZH
04-25.94	61155.94	RP	16.88	0.19	5.3	OBS-BZH
04-27.91	61157.91	G	17.48	0.09	11.0	OBS-BZH
04-27.93	61157.93	BP	17.23	0.07	14.2	OBS-BZH
04-27.95	61157.95	RP	16.74	0.12	8.2	OBS-BZH
05-01.91	61161.91	G	16.35	0.03	32.1	DAX T500

Color Term: G-band across sites



Zero Point per Night



Findings

- Unreliable first epoch** — On April 24, the template subtraction yields S/N = 2.88 (G) and 2.28 (RP), below the S/N = 3 detection threshold. The color index CI(BP - RP) = -0.37 is unphysical and entirely driven by noise in the RP measurement (err = 0.44 mag). The subsequent nights give CI = +0.60 ± 0.21 and +0.49 ± 0.14, consistent within uncertainties.
- Observation conditions** — April 24 had the poorest atmospheric transparency (lowest zero points in all bands, ~0.4 mag below the other nights), worse seeing (FWHM 4.7-5.2 px vs 4.1-4.4 on Apr 27), and fewer stacked frames (37 vs 60 for BP/RP), all degrading the subtraction S/N.
- No filter swap** — Photometric color terms are remarkably stable across all 3 nights: G ≈ -0.25, BP ≈ +0.03, RP ≈ -0.18 (spread < 0.02). A filter swap would manifest as a sudden jump in color term. Intrinsic calibration scatter = 0.01 mag on all 9 tasks.